sybil-dotfiles

How this machine is set up, and why

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1 Power and battery

The machine is a ThinkPad T480 (`20L6`): an i5-8350U rated at 15 W, Intel UHD 620 graphics, a 1080p panel, and two batteries — `BAT0` inside, `BAT1` the removable one at the back. Battery tuning is split across four owners, each doing the part only it can do:

OWNER	WHAT IT CONTROLS
TLP	Kernel and device knobs: CPU turbo and energy preference, PCIe ASPM, runtime PM, radios, charge thresholds.
throttled	The CPU package power limits (PL1/PL2), rewritten every few seconds because the T480's firmware keeps resetting them.
Hyprland	GPU work: blur, shadows, animations, panel refresh rate.
quickshell	The plug/unplug event that tells Hyprland and user services to switch profile.

1.1 Already in place

Before this round of changes the basics were covered:

- TLP with charge thresholds of 75/80 % on both batteries (`/etc/tlp.conf`), plus `tlp-rdw` for radio switching on NetworkManager events.
- throttled, which works around the T480's aggressive embedded-controller throttling on AC.
- zram swap (zstd, 7.7 GB) with zswap disabled on the kernel command line.
- S3 deep sleep selected (`/sys/power/mem_sleep` reads `s2idle [deep]`).
- hypridle dimming, locking and blanking the panel; `paccache.timer` and `fstrim.timer` .

1.2 CPU: power limits and turbo

1.2.1 throttled

The stock `/etc/throttled.conf` lets the CPU sustain 29 W on battery — about twice its rated 15 W TDP. The battery section now sets `PL1_Tdp_W: 15`; the AC section is unchanged at 44 W. PL2 (the short burst limit) stays at 44 W, so bursts still feel snappy.

> ⚠ **Warning (Undervolting)** BIOS `N24ET81W (1.56)` almost certainly locks the voltage MSRs (Lenovo's fix for Plundervolt, CVE-2019-11157). To check, set `CORE: -50` under `[UNDERVOLT.BATTERY]`, then read `journalctl -u throttled -n 30`. If it reports that undervolting is unsupported or locked, put it back to 0.

1.2.2 TLP drop-in

TLP was running on its defaults apart from the thresholds. `system/etc/tlp.d/10-battery.conf` adds:

SETTING	VALUE	EFFECT
`CPU_BOOST_ON_BAT`	`0`	No turbo: the CPU stays at or below its 1.7 GHz base clock. The largest single saving, at a real cost to single-threaded speed.
`CPU_HWP_DYN_BOOST_ON_BAT`	`0`	Stops the kernel raising the minimum P-state for I/O-bound tasks.
`PCIE_ASPM_ON_BAT`	`powersupersave`	Deepest PCIe link power states (the default leaves the kernel's choice). The NVMe's Silicon Motion SM2263 controller has a history of stalls in deep link states: if the disk hangs on battery, drop to `powersave`.
`DEVICES_TO_DISABLE_ON_BAT_NOT_IN_USE`	`bluetooth wwan`	Radios with nothing connected are switched off when the charger is pulled.
`CPU_MAX_PERF_ON_BAT`	*commented*	A further cap if turbo-off is not enough.

TLP's own battery defaults were already right and are left alone:

- `CPU_ENERGY_PERF_POLICY_ON_BAT=balance_power`
- `RUNTIME_PM_ON_BAT=auto`
- `WIFI_PWR_ON_BAT=on`
- `USB_AUTOSUSPEND=1`

> ※ **Remark** `/etc/modprobe.d/` sets `iwlwifi power_save=0`. That only sets the driver default; TLP still turns Wi-Fi power saving on when on battery. It was kept as is.

Because TLP now switches radios, the `systemd-rfkill` service and socket are masked, as TLP's documentation asks — otherwise the two fight over radio state at boot.

1.3 Compositor: the battery profile

On the UHD 620 the costliest thing Hyprland does is blur: 3 passes behind every inactive window, which are 75 % opaque, plus a 35 px shadow under each. The battery profile turns these off.

`hyprland.lua` reads `/sys/class/power_supply/AC/online` once each time it loads (Hyprland's Lua has `io`) and sets:

SETTING	ON AC	ON BATTERY
`decoration.blur.enabled`	on	off
`decoration.shadow.enabled`	on	off
`animations.enabled`	on	off
`eDP-1` mode	`preferred` (60 Hz)	`1920x1080@48`
`decoration.blur.xray`	on	on

`xray` is now on all the time. Tiled windows never overlap in the scrolling layout, so it only changes floating windows: they blur the cached wallpaper instead of re-blurring whatever is under them.

The panel's 48 Hz mode appears in `hyprctl monitors` under `availableModes`. It was tested at runtime (it reports `47.999 Hz`), and a reload restores 60 Hz.

! **Important (Why a reload, not hyprctl eval)** Switching profile is just `hyprctl reload`. If the profile were applied with `hyprctl eval` instead, the next config save (Hyprland reloads on save) would silently put AC settings back on battery. Doing it at load time makes `hyprland.lua` the only source of truth.

A reload does *not* re-run the `hyprland.start` hook (checked: still one quickshell and one miniserve after a reload), so the quickshell → reload → quickshell loop it could otherwise cause cannot happen.

1.4 The plug/unplug hook

`quickshell/Common/Battery.qml` watches UPower's `onBattery` property and, when it flips, runs `~/.config/hypr/scripts/powermode.sh`. The script:

1. runs `hyprctl reload`, so `hyprland.lua` picks the matching profile;
2. stops `syncthing.service` (a user unit) on battery and starts it again on AC. Set `PAUSE_SYNCTHING=0` at the top of the script to keep syncing on battery.

TLP and throttled detect the AC state on their own (TLP through udev, throttled by polling sysfs), so the hook does not touch them.

※ **Remark** The hook is meant for *changes*. If you log in on battery, Hyprland already starts in the battery profile (it reads the AC state at load). Whether syncthing stops depends on quickshell: UPower's properties fill in asynchronously, so `onBattery` may flip once at startup and run the hook, which stops syncthing; if it doesn't flip, syncthing runs until the next plug or unplug. Either way nothing breaks: a reload does not restart quickshell.

1.5 Idle: suspend on battery

hypridle's timeline, measured from the last input:

AFTER	ACTION
2.5 min	Dim the panel to 10 % (`scripts/idle-dim.sh`), keyboard backlight off
5 min	Lock (`loginctl lock-session`)
5.5 min	Panel off (DPMS)
15 min	**Suspend, on battery only**

The 15-minute listener checks `AC/online` when its timer fires, so AC keeps the old no-auto-suspend behaviour. Unplugging partway through an idle stretch does not trigger it until the next idle stretch. Deep sleep (S3) is already selected, so a suspended T480 costs very little. quickshell's keep-awake toggle still pauses every timer.

1.6 Video decoding

The GPU decodes H.264, HEVC and VP9 in hardware, but not AV1. YouTube prefers AV1, which then gets decoded on the CPU — one of the most expensive things a laptop can do on battery.

- `hyprland.lua` sets `LIBVA_DRIVER_NAME=iHD`. Both `intel-media-driver` (iHD) and `libva-intel-driver` (i965) are installed, and this makes the choice explicit.
- Firefox gets `media.av1.enabled = false` from a `user.js` in the default profile (`ee392ds5.default-release`). That file is not in this repo: Firefox profiles are not stowed. YouTube then falls back to VP9.

To check, play a video and open `about:support` → *Media*, or run `vainfo` (from `libva-utils`) and look for `iHD` and the `VAProfileVP9` entries.

1.7 Packages

The new `system.power` feature in `deps.toml` owns the power stack, and `cli.monitoring` is now on, with `btop` in place of `bpytop`. Both are in the `desktop` profile.

PACKAGE	FEATURE	WHY
`tlp` , `tlp-rdw`	`system.power`	Already installed; now declared.
`throttled`	`system.power`	Already installed; now declared.
`powertop`	`system.power`	*Measuring only.* Its `--auto-tune` fights TLP.
`libva-utils`	`system.power`	`vainfo` , to confirm hardware decoding.
`intel-gpu-tools`	`system.power`	`intel_gpu_top` : see what blur and animations cost.
`nvme-cli`	`system.power`	NVMe power states (APST) and health.
`btop` , `duf`	`cli.monitoring`	Process, battery and disk monitors.

1.8 Installing

Through the manager, `system.power` is in the `desktop` profile; its setup steps (`src/tui/features/system_power.py`) copy the `/etc` files. stow only links into `$HOME`, so root-owned files live in `system/etc/` and are copied rather than linked. By hand, from the repo root:

```
sudo pacman -S --needed powertop libva-utils intel-gpu-tools nvme-cli btop duf
sudo systemctl mask systemd-rfkill.service systemd-rfkill.socket
sudo install -Dm644 system/etc/tlp.d/10-battery.conf /etc/tlp.d/10-battery.conf
sudo tlp start
sudo install -Dm644 system/etc/throttled.conf /etc/throttled.conf
sudo systemctl restart throttled
```

> ⚠ **Warning** `system/etc/throttled.conf` replaces the whole file. Edit the repo copy and reinstall it — edits made only in `/etc` are lost on the next install.

1.9 Checking it works

Unplug the charger, wait a few seconds, then:

```
tlp-stat -c | grep 10-battery          # the drop-in is being read
sudo tlp-stat -p                       # no_turbo = 1, EPP balance_power
journalctl -u throttled -n 20          # PL1 15 W applied on battery
hyprctl getoption decoration:blur:enabled   # false
hyprctl monitors | grep -A1 eDP-1      # 1920x1080@47.999
systemctl --user is-active syncthing   # inactive
upower -i /org/freedesktop/UPower/devices/battery_BAT1 | grep energy-rate
```

`energy-rate` (or `sudo powertop`'s overview) is the number to watch: compare it before and after a change, with the same brightness and workload.

`sudo cat /sys/kernel/debug/dri/*/i915_edp_psr_status` shows whether the panel uses Panel Self Refresh. That could not be checked without root.

1.10 Battery health

Read on 2026-10-04:

BATTERY	DESIGN	FULL NOW	HEALTH	CYCLES
`BAT0` (internal)	24.0 Wh	15.4 Wh	64 %	416
`BAT1` (external)	71.0 Wh	56.0 Wh	79 %	352

At the time `BAT0` was at 6 % on AC while `BAT1` charged at 22.9 W. That may just be the order the firmware charges in, but run `sudo tlp-stat -b` to check that `BAT0` does climb back up.

With 75/80 % thresholds the batteries never fill. Before a long day away from power, run `sudo tlp fullcharge` to fill to 100 % (it lasts until the next boot, when the configured thresholds return), or `sudo tlp fullcharge BAT1` to fill just the external battery.

1.11 Left out, on purpose

- **power-profiles-daemon, auto-cpufreq, tuned:** each conflicts with TLP. **thermald** conflicts with throttled.
- `powertop --auto-tune`: it overwrites TLP's settings until reboot.

- **systemd-oomd:** not power related. On Arch no slice has `ManagedOOM*` set, so enabling the service does nothing without extra drop-ins. With 16 GB RAM plus zram it is not worth it.
- **hyprsunset:** no config uses it, and the manifest only lists packages something needs.
- **Tailscale down on battery:** idle `tailscaled` is cheap, and taking it down from a user script needs `tailscale set --operator=$USER`. Do it by hand when it matters: `sudo tailscale down`.

1.12 Files

PATH	ROLE
`system/etc/tlp.d/10-battery.conf`	TLP battery drop-in → `/etc/tlp.d/`
`system/etc/throttled.conf`	Full throttled config → `/etc/`
`configs/hypr/…/hyprland.lua`	`onBattery` at load; battery profile; `LIBVA_DRIVER_NAME`
`configs/hypr/…/scripts/powermode.sh`	Reload Hyprland; pause or resume syncthing
`configs/hypr/…/hypridle.conf`	Suspend after 15 min, on battery only
`configs/quickshell/…/Common/Battery.qml`	Runs `powermode.sh` when `UPower.onBattery` changes
`deps.toml`	`system.power` feature; `cli.monitoring` on, `btop`
`src/tui/features/system_power.py`	Copies the `/etc` files, masks rfkill
Firefox profile `user.js`	`media.av1.enabled = false` (not in this repo)